

SECTION BAnswer **all** questions.

6. Ammonia is a versatile compound produced on a large scale. It is used in fertilisers and cleaning products and to make a range of other nitrogen-containing compounds.

(a) Ammonia is an example of a weak base.

- (i) State what is meant by a base and explain why the ammonia molecule is able to act as a base. [2]

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(ii) Ammonia can be used as part of a mixture that forms a buffer.

- I. Suggest a compound that could be added to ammonia solution to form a buffer. [1]

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- II. Suggest a use for a buffer. [1]

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(b) Ammonia can form a compound with borane (BH_3).

- (i) Draw a dot and cross diagram of the compound formed. [1]



- (ii) Boron-nitrogen bonds are also present in boron nitride. One form of boron nitride (hexagonal boron nitride) is sometimes called white graphite.

Complete the table below giving similarities and differences in the structure and bonding of hexagonal boron nitride and graphite. One similarity has been included for you. [2]

	Similarity between hexagonal boron nitride and graphite	Difference between hexagonal boron nitride and graphite
Structure		
Bonding	each atom is bonded by covalent bonds to three others in both graphite and BN	

- (c) Nitrogen forms a range of compounds such as NH_3 and NF_3 which have three covalent bonds. Phosphorus can form compounds such as PF_3 which has three covalent bonds and PF_5 which has five covalent bonds.

Explain this difference in the chemistry of nitrogen and phosphorus. [2]

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